To Renounce Liberty? A Public-Goods Experiment on (Un)Democratic Reversal

Hector Bahamonde[*1,2], Inga Saikkonen[*2], Janne Tukiainen[*3], and Åsa von Schoultz[*4]

[1]INVEST Research Flagship Centre, University of Turku, Finland
[2]Social Science Research Institute, Åbo Akademi University, Finland
[3]Turku School of Economics, University of Turku, Finland
[4]Faculty of Social Sciences, University of Helsinki, Finland

hector.bahamonde@utu.fi inga.saikkonen@abo.fi
janne.tukiainen@utu.fi asa.vonschoultz@helsinki.fi

Abstract

Democratic backsliding often proceeds through lawful choices that weaken constraints on later decisions. Yet we know far more about why citizens surrender institutional safeguards than about whether they reclaim them once the crisis that justified concentrated authority recedes. We studied this reverse movement in an incentivized public-goods experiment with 300 participants in Finland. Citizens repeatedly decided whether an unknown leader's proposal required group approval or took effect automatically. Automatic implementation initially offered higher productivity and assured coordination, making weaker constraints materially attractive. We then randomly assigned matching pools to continued crisis or structural recovery, which made approval equally productive while leaving the leader's authority unchanged. The central outcome was whether citizens who had authorized decisions without approval subsequently voted to restore it. The design treated democratic reversal as a consequential choice over decision rights rather than a declaration of democratic sentiment. It therefore identified whether structural recovery caused citizens to withdraw the support that sustains executive aggrandizement and whether democratically authorized losses of decision rights remained reversible.

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[*]All authors are listed in alphabetical order and contributed equally to the paper.

1 Research Question and Contribution

Democratic backsliding depends not only on leaders who weaken constraints but also on citizens who authorize those changes. One common form is *executive aggrandizement*, in which formally lawful decisions gradually reduce constraints on one-person authority. This study examined its demand-side microfoundation by asking whether citizens who surrendered an approval right later voted to restore it after a crisis receded. We define that within-person change in institutional support as **(*un*)democratic reversal**.

We build on the democratic backsliding and public-good games literatures. Contemporary backsliding often advances without ending elections because citizens may support elected officials or lawful rule changes that reduce the approval, oversight, or veto constraints governing later decisions (Bermeo 2016). The politically relevant step is therefore not necessarily an explicit vote against democracy. It can instead be a *democratic* vote that makes subsequent authority *less* constrained. Threat and policy benefits can increase tolerance for weaker constraints (Hetherington and Suhay 2011; Goodman 2021; Saikkonen and Christensen 2023). Laboratory evidence similarly shows that gridlock can increase support for executive special powers (Forteza, Mussio, and Pereyra 2024).

This literature mainly explains movement toward less constrained authority. We followed the reverse movement by observing whether the same citizens restored an approval right after its crisis-based benefit receded. That second choice turned reversal into a consequential decision over an institutional constraint rather than a stated evaluation of democracy.

Public-good games supply an economic reason for weakening approval. Voluntary provision is vulnerable to free riding, which means keeping one's points while others fund the shared service. This individual incentive can erode cooperation, whereas centralized enforcement can coordinate contributions (Fehr and Gächter 2000; Fischbacher and Gächter 2010; Hamman, Weber, and Woon 2011). Experiments also show that voting over institutions changes their performance (Dal Bó, Foster, and Putterman 2010; Sutter, Haigner, and Kocher 2010; Markussen, Putterman, and Tyran 2014). We joined these literatures by making citizen approval both institutionally meaningful and economically costly.

In each round, five citizens began with 20 points each and voted between two rules for managing a shared public-services fund. Under both rules, the software selected one citizen uniformly at random after the vote. Each citizen therefore had the same one-in-five chance of leading that round. The leader proposed how many points every citizen placed in the fund and whether any collected points moved to the leader's payoff. The game multiplied the remaining fund points and divided the group return equally.

Under the approval-constrained rule (D), at least three of the other four citizens had to approve the complete proposal. Rejection returned the group to voluntary individual allocations. Under automatic implementation (A), the same proposal took effect without that approval vote. Repeated interaction occurred inside a stable *matching pool* of 10 or 15 citizens. Before each round, the software reshuffled that pool into two or three temporary groups of five. The institutional ballot therefore selected an approval rule, not a known leader or proposal.

Everyone began with a public-service crisis. Participants read that "*Parliament has failed to pass a budget for eight months, healthcare waiting times have doubled, and planning for public services has stalled.*" During Block 1, each fund point created 1.50 group points under D and 2.50 under A. Direct implementation therefore had a one-point *productivity premium* and guaranteed that the leader's proposal was implemented. After Block 1, each matching pool received **institutional recovery** or **institutional persistence**. Recovery combined improved public-service conditions with a rise in the D rate to 2.50. Persistence retained the crisis and the 1.50 rate. The A rate and decision rule remained fixed.

The payoffs made the choice consequential. **The *allocation of decision rights* connected it to democratic backsliding**. A citizen who voted for A supported surrendering an approval vote before knowing the leader or proposal. A later vote for D supported restoring that safeguard after the crisis-based productivity advantage disappeared. *The experiment therefore observed whether citizens voted to take back a specific constraint on authority rather than inferring reversal from a survey declaration.*

In simple terms

Five citizens voted before seeing who would lead or what the leader would propose. Under D, three of the other four citizens had to approve the proposal. Rejection returned everyone to individual fund allocations. Under A, the proposal took effect directly. In Block 1, these methods used rates of 1.50 and 2.50, respectively.
After Block 1, recovery improved public-service conditions and raised D to 2.50. Persistence left the crisis and rate gap in place. Among previous A supporters, a later D vote supported restoring the approval right. Comparing recovery with persistence identified how much the structural recovery package changed that reversal.

Central insight

A decision can be democratically authorized while weakening the democratic constraints that govern what happens next. This resembles contemporary backsliding, which often proceeds through elections and formally lawful institutional changes rather than an outright abolition of democracy (Bermeo 2016).
In the experiment, the first majority vote was democratic. A citizen who chose A voted to let an unknown leader's later allocation and transfer take effect without the other citizens' approval. That vote removed a specific constraint while leaving the institutional ballot itself intact. A later vote for D supported restoring the approval right that the citizen had previously voted to surrender.

What matters is that citizens used one democratic decision to weaken their control over a subsequent decision and could later vote to take that control back.

2 Theory

2.1 Crisis and the productivity premium

A crisis can make action without approval attractive when ordinary procedures produce delay or failed coordination. The experiment gave this argument material consequences. In Block 1, proposals requiring approval (D) used a 1.50 fund rate, while proposals implemented directly (A) used 2.50. Under D, rejection also returned the group to voluntary contributions and possible free riding. Direct implementation therefore offered both a productivity premium and assured implementation of the leader's equal allocation (Fehr and Gächter 2000; Fischbacher and Gächter 2010; Hamman, Weber, and Woon 2011).

The personal-transfer option made the institutional constraint consequential. The leader could propose moving up to 20 fund points to their own payoff under either procedure. Under D, the other citizens saw that transfer before implementation and could reject it. Under A, they observed it only after it had taken effect. The experiment therefore concerned approval over discretion, not

the prevalence of corruption in Finland.

H1: Authorization under crisis. During Block 1, support for direct implementation (A) increases when its productivity and coordination advantages produce better outcomes. Larger observed transfers reduce subsequent support.

2.2 Citizen authorization and democratic constraint

The institutional ballot occurred before leader selection and before the proposal. A vote for A therefore authorized a rule rather than a known person or action. The citizen relinquished participation in approving the next binding decision, whereas a vote for D retained that right. **The design thus provided an abstract microfoundation of *executive aggrandizement* in which a *democratic* choice *weakened* the constraint governing what followed** (Bermeo 2016; Forteza, Mussio, and Pereyra 2024).

H2: Approval and discretion. Approval declines as proposed personal transfers increase. Anticipating rejection, leaders operating under D propose smaller transfers than leaders whose proposals take effect directly. Because groups choose the institutional rule themselves, these patterns describe behavior within the chosen rules rather than a separate randomized effect.

2.3 Recovery and democratic reversal

Recovery was assigned at the matching-pool level. Pool g contained 10 or 15 participants who were reshuffled into two or three temporary groups of five before each round. Three votes produced a majority within each group. The entire pool received the same Block-2 condition. We use $T_g = 1$ for recovery and $T_g = 0$ for persistence. Let m_{gb} denote the approval-constrained fund multiplier in pool g and block b. The productivity premium for direct implementation is

$$\delta_{gb} = 2.50 - m_{gb} \tag{1}$$

Everyone faced $\delta_{g1} = 2.50 - 1.50 = 1$ in Block 1. Recovery was implemented as one structural package. Participants learned that public-service conditions had improved, and m_{g2} rose to 2.50, which set $\delta_{g2} = 0$. Persistence reported that the crisis continued and kept $m_{g2} = 1.50$, so $\delta_{g2} = 1$. The leader-selection rule, proposal set, approval threshold, rejection-to-voluntary-allocation rule, and A rate remained fixed.

The recovery–persistence comparison identified the total causal effect of structural recovery on (un)democratic reversal. Among final-round Block-1 supporters of A, the analysis compared later D votes in recovery pools ($T_g = 1, \delta_{g2} = 0$) with those in persistence pools ($T_g = 0, \delta_{g2} = 1$). The public-service description stated what changed, and the production rate made that change materially consequential inside the game. Together, they defined the recovery package. The estimand therefore captured their combined effect rather than separate framing and payoff effects. Persistence supplied the counterfactual in which both the crisis and premium remained.

H3: Immediate restoration. Among participants who vote for A in Block 1's final round, recovery increases the probability of voting for D in Block 2's first round.

H4: Sustained reversal. Among the same participants, recovery increases the probability of voting for D in Block 2's final round.

3 Experimental Design

3.1 Overview

Each laboratory cohort completed one session of approximately 60 minutes that contained two ten-round blocks. The University of Turku issued a favorable ethical-review statement on 30 September 2026 (case TY/1722/2026). Sessions took place at the UTU Choice Lab decision-making laboratory, which has 24 regular seats and one accessible seat. We capped production rosters at 20. Participants remained in matching pools of 10 or 15 and were reshuffled into anonymous five-person groups before every paid round. After Block 1, each entire pool was assigned to recovery or persistence. The pool then learned whether public-service conditions had improved or remained strained and saw its Block-2 fund rates. This structure created repeated interaction without stable five-person teams. In the figures, p_i identifies participant i.

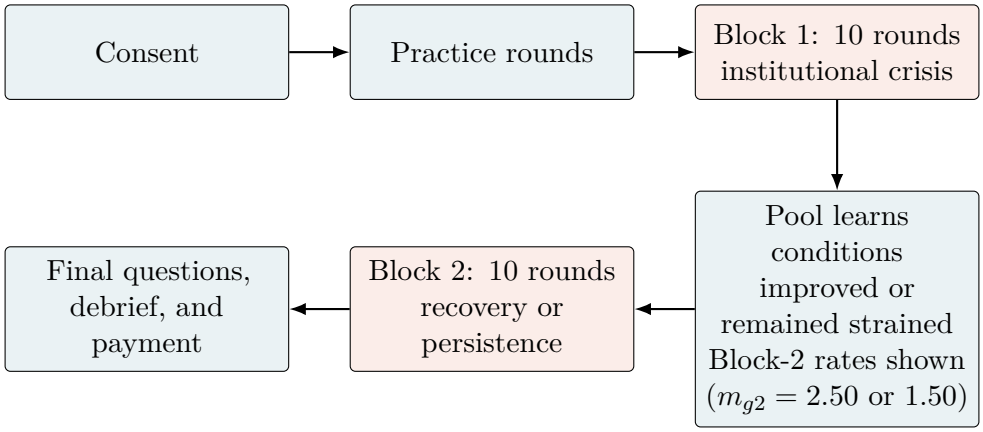


Figure 1: One-session sequence. Each paid round included an institutional ballot, leader proposal, implementation, feedback, and rematching.

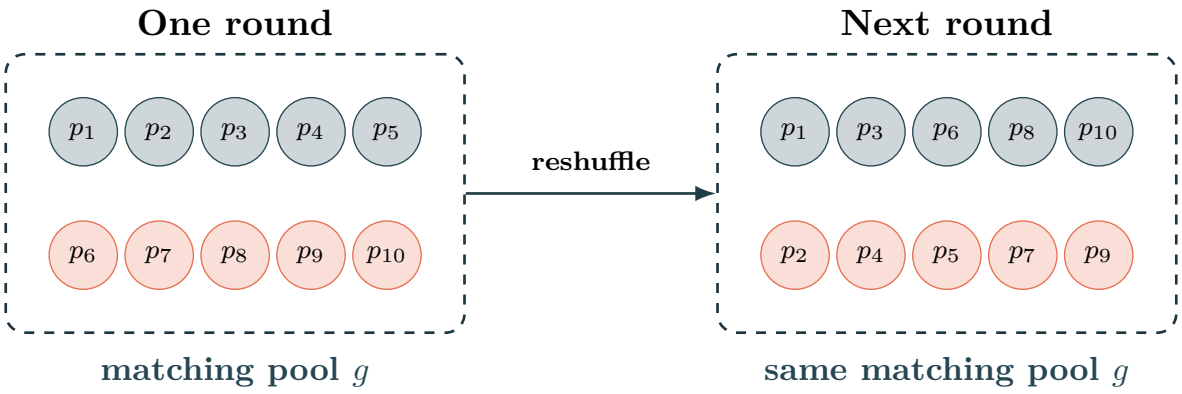


Figure 2: Anonymous rematching within a matching pool (ten-person example).
Notes: Each participant p_i remained in pool g. A ten-person pool formed two temporary groups per round, while a fifteen-person pool formed three. Recovery or persistence applied to the whole pool in Block 2.

In simple terms

Matching pools permitted interaction without fixing the same five-person team. Round-by-round reshuffling limited stable reputations, collusion, and retaliation while keeping participants inside one treatment environment. We used ten-person pools whenever the completed cohort could be divided that way. One fifteen-person pool accommodated a remaining set of five while retaining rematching.

Reversal was measured for individuals, but treatment (T_g) was assigned to pools. We therefore clustered statistical uncertainty by pool. Temporary groups organized play but were not treatment units. Ten-person pools remained the default because they yielded more independent treatment assignments for a fixed sample.

Table 1: Session formation from the completed laboratory roster

People who arrive	Matching pools	Five-person groups per round	Participate	Released
20	$10 + 10$	$2 + 2 = 4$	20	0
16–19	15	3	15	1–4
15	15	3	15	0
11–14	10	2	10	1–4
10	10	2	10	0
Fewer than 10	—	—	0	All

Notes: A matching pool is not a playing group. Each ten-person pool formed two temporary five-person groups per round, and a fifteen-person pool formed three. At 20 participants, two separate ten-person pools each formed two groups. Groups dissolved and were reshuffled after every round. Participants never entered or changed pools after paid play began.

3.2 The institutional-choice public-good game

Each citizen started with 20 points and privately voted between an approval-constrained rule (D) and automatic implementation (A). Let $V_{i,b,r} \in \{\mathsf{D}, \mathsf{A}\}$ denote citizen i's ballot in block b and round r. Three votes selected the binding rule. The software then selected each group's leader through an independent uniform draw among its five citizens. Selection occurred under both rules and only after the institutional ballot. Every citizen therefore had the same chance before the draw in every round, and past service did not predict the next draw.

Table 2: Implemented design parameters

Feature	Implemented value
Laboratory session	One synchronized visit, approximately 60 minutes
Paid rounds	10 per block (20 total), preceded by unpaid practice
Group and matching pool	5-person groups within stable 10- or 15-person matching pools
Rematching	Anonymous rematching within pool before every paid round
Round endowment	20 points per participant
Approval-constrained multiplier	1.50 in Block 1 and 2.50 under recovery or 1.50 under persistence in Block 2
Automatic-implementation multiplier	2.50 in both blocks and conditions
Approval	At least 3 of the 4 nonleaders, with a 2–2 split rejecting the proposal
Rejection	Voluntary individual contributions at the approval-constrained rate
Personal transfer	0–20 fund points, limited by the points collected
Performance bonus	Points earned in one randomly selected round from each block
Maximum performance bonus	EUR 5.00
Show-up payment	EUR 10.00
Maximum total payment	EUR 15.00

Selected-round points are converted linearly into money. Because a selected round can yield at most 60 points, the two selected rounds produce a bonus of at most EUR 5. Selecting one round from each block keeps every decision consequential without paying all rounds or encouraging participants to offset one decision with another.

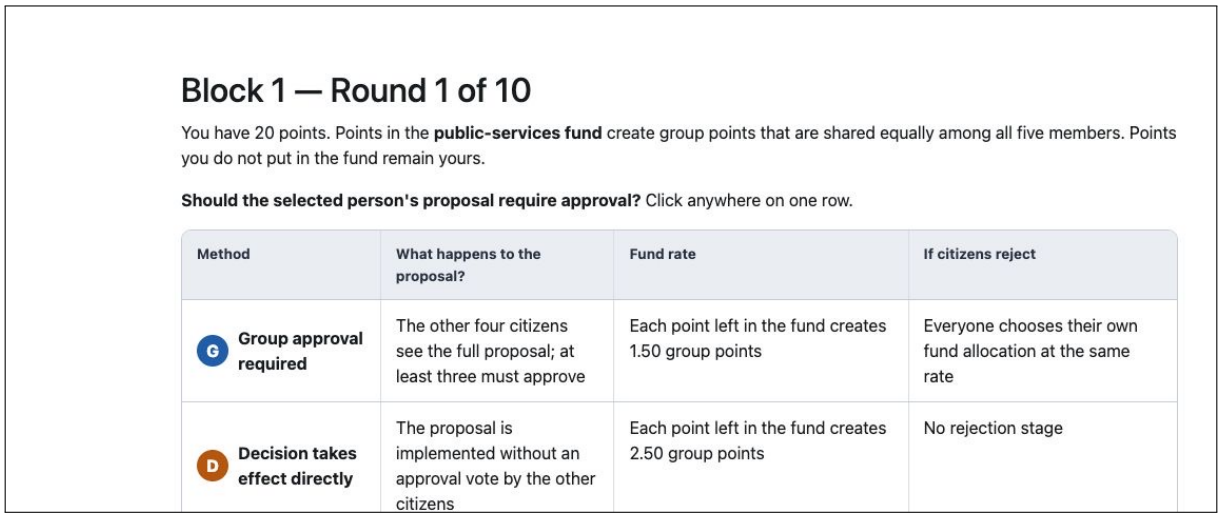


Figure 3: Participant screen for choosing whether the leader's proposal requires approval.

Leader proposal. Under either rule, the leader proposed an equal allocation $t \in \{0, \ldots, 20\}$ and a personal transfer $\rho \in \{0, \ldots, 20\}$, with $\rho \leq 5t$. Let π denote a citizen's payoff in that round.

Under A, the proposal took effect directly, and a nonleader earned

$$\pi^{A}_{igbr} = 20 - t + \frac{2.50}{5}(5t - \rho), \tag{2}$$

and the leader earned $\pi^{A,L}_{igbr} = \pi^{A}_{igbr} + \rho$.

Approval-constrained procedure (D). The four nonleaders observed (t, ρ) and voted privately. Let $q_{igbr} = 1$ denote approval. The proposal passed when at least three approved. If it passed, the payoffs above applied with m_{gb} replacing 2.50. If it failed, each citizen chose $c_{igbr} \in \{0, \ldots, 20\}$ and earned

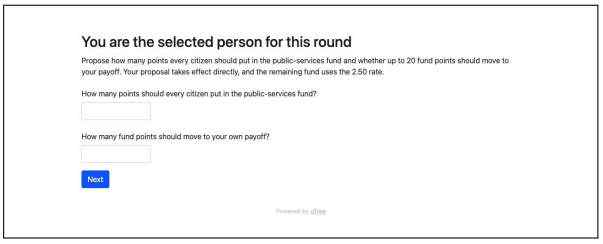
$$\pi^{D}_{igbr} = 20 - c_{igbr} + \frac{m_{gb}}{5} \sum_{j=1}^{5} c_{jgbr}. \tag{3}$$

The rejected transfer was not implemented. Everyone then saw the proposal, approval total, implemented allocation, transfer, and own payoff.

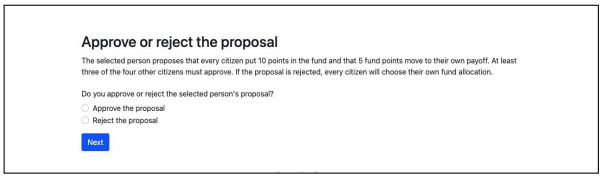
In simple terms

Rejection restored the standard **public-goods dilemma**. If one citizen contributed 1 point and four contributed 0, the contributor earned $19 + \frac{1.50}{5} = 19.30$ during the crisis, while each free-rider earned 20.30. Under recovery, these payoffs became 19.50 and 20.50. Contributing helped the group but cost the contributor. Direct implementation avoided that risk, whereas approval allowed citizens to block an unwanted proposal.

(a) Leader proposal



(b) Citizen approval



(c) Proposal, approval, and payoff feedback

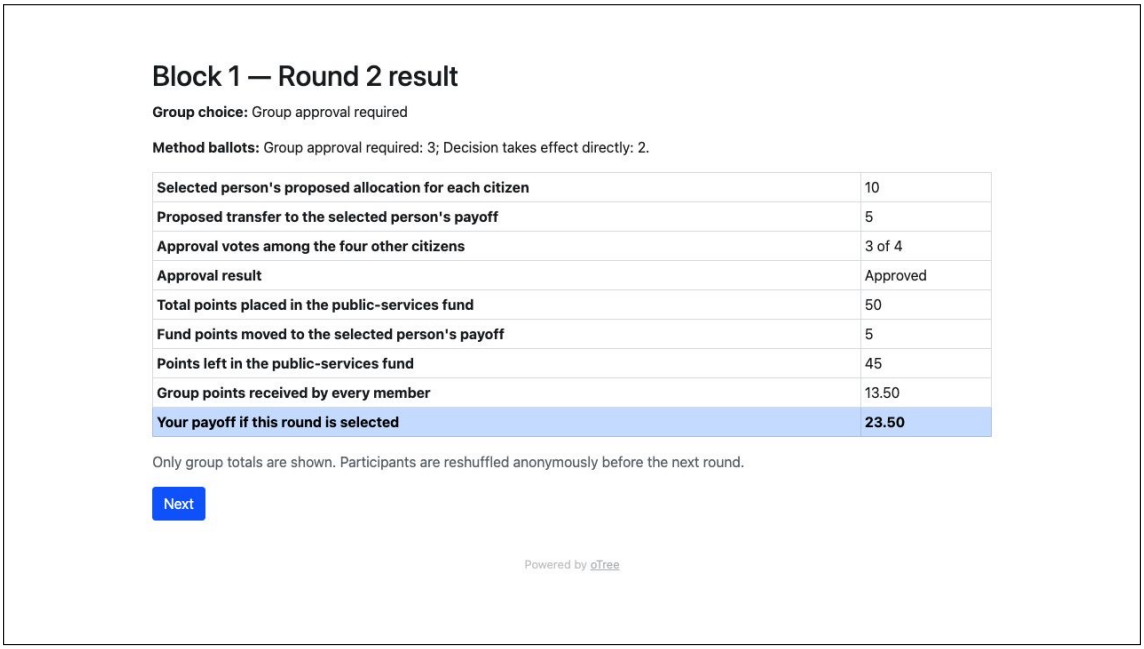


Figure 4: Proposal, approval, and round feedback. Entered amounts are illustrative.

How incentives map onto reversal. During Block 1, A combined a 2.50 rate with guaranteed implementation. By contrast, D used 1.50 and could return to voluntary contributions after rejection. Recovery reported improved public-service conditions and raised D to 2.50, while persistence retained the crisis and rate gap. The proposal set, approval rule, rejection procedure, and A rule remained unchanged. The recovery–persistence contrast therefore estimated the total effect of this structural package.

From institutional choice to (un)democratic reversal. A citizen who moved from A to D first voted to let an unknown leader's proposal take effect without approval and later voted to restore that approval. The public-good incentives made this act consequential. After recovery, A still guaranteed implementation of the leader's equal allocation, whereas D could produce rejection and free riding. A higher switching rate under recovery would therefore indicate that structural change caused citizens to withdraw support for a weakly constrained decision rule. *Reversal was observed as a vote to restore a decision right rather than as a declaration of democratic sentiment.*

A simple payoff comparison

Suppose the leader proposed $t = 10$ and $\rho = 0$, and the proposal was approved under D. Holding that decision fixed gave the following payoffs.

	Approval required (D)	Direct implementation (A)
Crisis or persistence	25 points each	35 points each
Recovery	35 points each	35 points each

Recovery made an approved D proposal equally productive, but it did not make that proposal pay more. A still guaranteed implementation. **The question was whether citizens voted to restore approval once it became equally productive.**

Participants reported expected payoffs and transfers before the final Block-1 ballot and the first and final Block-2 ballots. Expected payoffs assessed whether they understood the rules' relative performance. Expected transfers traced beliefs about how leaders would use their discretion.

In simple terms

Five citizens cast private institutional ballots ($V_{i,b,r}$), after which the software selected a leader. That leader proposed (t, ρ). Under D, three of four nonleaders had to approve. Rejection activated individual contributions (c_{igbr}). Under A, the proposal took effect directly. Everyone saw the result and payoff before rematching within pool g.

3.3 Block 1: crisis and leader authorization

After consent, participants practiced proposing, approving, rejecting, and making individual allocations after rejection. They then answered five comprehension questions. The interface never called either method democratic or undemocratic.

Block 1 began with the public-service crisis described above. Each fund point produced 1.50 under D and 2.50 under A. The leader could propose the same allocation and transfer under either rule. The rules differed in whether the other citizens could reject that proposal.

Each paid round repeated the same sequence. Citizens cast an institutional ballot, the leader made a proposal, and the proposal either faced approval or took effect directly. Rejection activated individual allocations before feedback and rematching. We elicited expectations before the final ballot and selected one Block-1 round for payment.

3.4 Block 2: structural change and reversal

After Block 1, pools were assigned to *recovery* or *persistence*. Recovery reported that the budget had passed, waiting times were falling, and planning had resumed. It also raised the D multiplier from 1.50 to 2.50. Persistence retained the crisis and the 1.50 rate. The A rate, leader selection, proposal set, approval threshold, and rejection procedure remained unchanged. Together, the updated description and rate formed the structural-recovery treatment.

The first Block-2 ballot measured immediate support for restoring approval. The final ballot measured sustained reversal and served as the primary outcome. We elicited expectations before

both ballots and asked about democratic constraints only after paid decisions. One Block-2 round was selected for payment.

4 Outcomes, Process Measures, and Estimands

An *estimand* is the population-level quantity that an analysis seeks to recover. We first define the individual choices that compose the primary estimand.

Let $\mathbb{1}\{\cdot\}$ equal 1 when the condition inside braces is true and 0 otherwise. The final Block-1 ballot defines endorsement of automatic implementation as

$$A_{i1} = \mathbb{1}\{V_{i,1,10} = \mathsf{A}\}. \tag{4}$$

For the theoretically relevant subgroup with $A_{i1} = 1$, reversal is

$$R_i = \mathbb{1}\{V_{i,2,10} = \mathsf{D}\}. \tag{5}$$

The primary estimand is the average treatment effect of structural recovery among Block-1 supporters of automatic implementation:

$$\tau_R = \mathbb{E}[R_i(1) - R_i(0) \mid A_{i1} = 1], \tag{6}$$

where $R_i(1)$ and $R_i(0)$ are participant i's potential outcomes under recovery and persistence. Recovery includes both improved public-service conditions and the 2.50 D rate, so τ_R is the total effect of that package. Because A_{i1} was measured before assignment, subgroup membership was fixed before treatment. The primary test repeatedly reassigned recovery according to the prespecified pool-level procedure and evaluated how unusual the observed contrast was. Because τ_R describes the average eligible citizen, the primary estimate weighted eligible participants equally. A secondary analysis gave each pool equal weight. Regression estimates clustered uncertainty by pool and included a prespecified indicator for fifteen-person pools.

Immediate reversal is

$$R_i^I = \mathbb{1}\{V_{i,2,1} = \mathsf{D}\} \quad \text{for } A_{i1} = 1. \tag{7}$$

The same comparison applied to R_i^I estimated immediate support for restoration. The final-round outcome R_i remained primary because it recorded whether that choice persisted.

In simple terms

The primary estimand (τ_R) asks how much recovery changed sustained reversal among citizens who had voted for direct implementation in Block 1's final round. These citizens satisfy $V_{i,1,10} = \mathsf{A}$ and $A_{i1} = 1$. Sustained reversal means voting for group approval in Block 2's final round, so $V_{i,2,10} = \mathsf{D}$ and $R_i = 1$. We compared this outcome under recovery ($T_g = 1$) and persistence ($T_g = 0$). Only one potential outcome was observed per citizen, so random assignment identified the difference. Assignment and inference operated at pool level.

For each leader i, t_{igbr} is the proposed allocation and ρ_{igbr} the proposed transfer. A transfer proposal is $M_{igbr} = \mathbb{1}\{\rho_{igbr} > 0\}$, with intensity $\rho_{igbr}/(5t_{igbr})$ when $t_{igbr} > 0$. Thus, $t = 10$ and $\rho = 5$ proposes moving 10% of the 50 collected points. Approval ballots q_{igbr}, proposal passage, contributions after rejection, and subsequent institutional ballots provide descriptive process evidence. Because groups choose their rule, these measures do not identify separate causal effects of recovery.

Immediate reversal R_i^I recorded the first response to the structural package before Block-2 interaction (see Figure 5). The primary outcome R_i recorded Round 10 after nine completed rounds. Secondary outcomes included late-round automatic-implementation support and realized group-level restoration.

Expected payoffs assessed whether participants understood how recovery changed the rules' relative performance, while expected transfers traced beliefs about leader discretion. After all paid choices, eleven process items measured perceived crisis severity and recovery, expected institutional performance, and views on approval, equal allocation, and transfers. A separate optional twelve-item background module recorded age group, gender, education, household financial strain, party voted for in Finland's 2023 parliamentary election, left–right placement, whether democracy is preferable to other forms of government, satisfaction with democracy in Finland, evaluation of a strong leader unconstrained by parliament and elections, external political efficacy, generalized trust, and willingness to take risks. These prespecified measures captured partisan and ideological trade-offs, distinguished principled democratic support from evaluations of democratic performance, and benchmarked cooperative and risk-related predispositions (Graham and Svolik 2020; Saikkonen and Christensen 2023; Claassen 2020; Dohmen et al. 2011; Glaeser et al. 2000). We collected them only after paid decisions. They supported precision and descriptive heterogeneity analyses but did not alter the randomized estimand or independently identify a treatment mechanism.

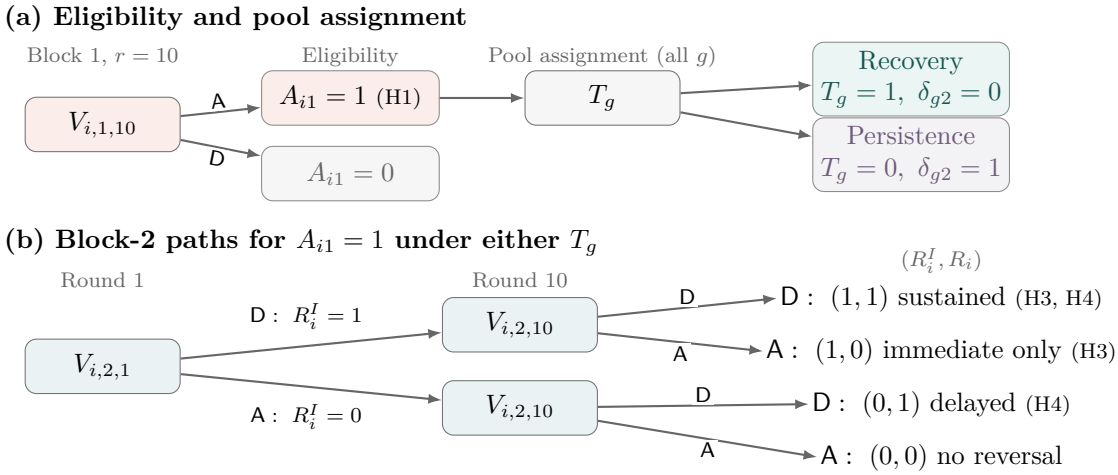


Figure 5: Decision tree for assignment and reversal. Each pool received one value of T_g. The primary analysis followed participants with $A_{i1} = 1$.

Notes: The displayed ballots are $V_{i,1,10}$ (final Block 1), $V_{i,2,1}$ (first Block 2), and $V_{i,2,10}$ (final Block 2). The ordered pair (R_i^I, R_i) records the two Block-2 outcomes, with $1 = \mathsf{D}$ and $0 = \mathsf{A}$. Thus, $(1, 1)$ is sustained reversal, $(1, 0)$ is immediate reversal only, $(0, 1)$ is delayed reversal, and $(0, 0)$ is no reversal. Parentheses identify the corresponding hypotheses. H2 concerns transfers and is not shown.

5 Randomization, Sample, and Design Sensitivity

The study included 300 completed participants in 28–30 matching pools. Ten-person pools were the default, and no more than four fifteen-person pools were used. This structure preserved at least 28 independent treatment assignments. Treatment was assigned only after everyone finished Block 1. Within each session, pools were ordered by their final-round Block-1 A vote share and paired. One pool in each pair received recovery, and the other received persistence. An unmatched pool was assigned by an independent fair draw. Because final-round A support was measured before

treatment, pairing balanced the number of participants eligible to reverse without conditioning on a post-treatment outcome. The maximum payment of EUR 15 set the participant-compensation ceiling at EUR 4,500.

The number of pools determined the amount of independent treatment information. The 300 participants produced 28–30 pool assignments, with approximately 14–15 pools per condition. A larger pool supplied more individual outcomes but still received only one treatment draw. Repeated rounds improved measurement and revealed trajectories, but they did not create additional independent assignments.

We evaluated design sensitivity with a simulation that reproduced pool-level treatment assignment, the eligible subgroup, the binary outcome, and the two-sided randomization test at the five-percent level. *Power* is the probability that this test detects a genuine recovery effect of an assumed size. It depends on three quantities that were unknown before the pilot. These were the share of participants eligible for the primary outcome, the sustained-reversal rate under persistence, and the *intrapool correlation*, which measures how similar outcomes are among members of the same matching pool. A treatment effect of 20 percentage points, for example, means a sustained-reversal rate 0.20 higher under recovery than under persistence.

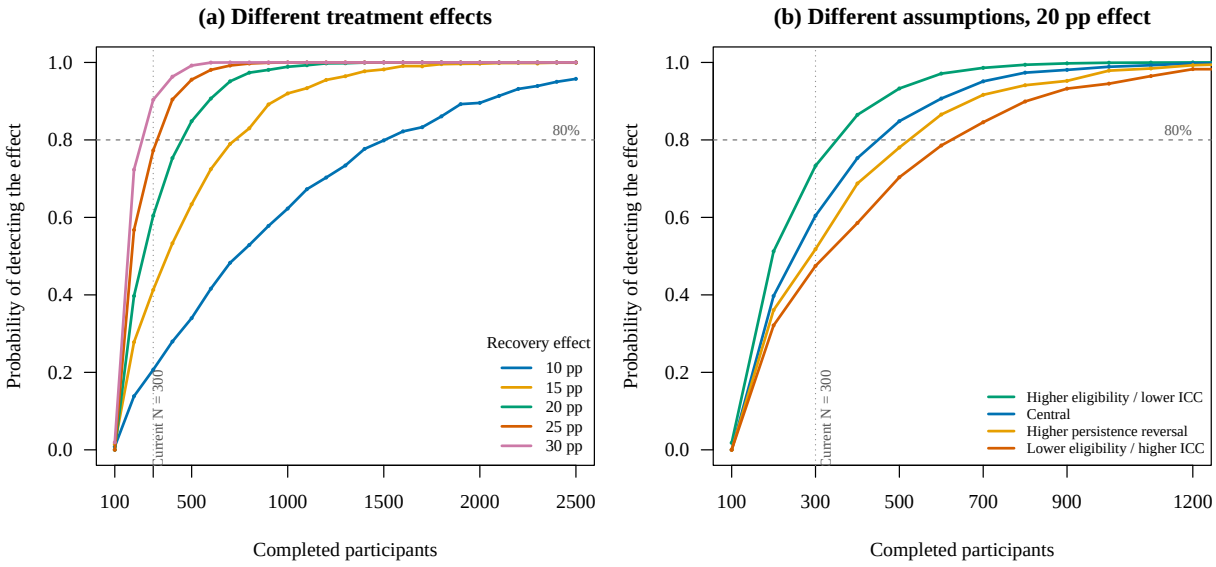


Figure 6: Simulated power as the completed sample increases

Notes: Panel (a) holds the eligible share at 0.50, the persistence reversal rate at 0.20, and the intrapool correlation at 0.05, then varies the true recovery effect. Panel (b) fixes that effect at 20 percentage points and varies the three uncertain parameters. Every additional 20 participants adds two ten-person pools and one paired recovery–persistence assignment. Points are simulation estimates and lines are monotone summaries. The horizontal line marks 80% power and the vertical line marks the planned sample of 300. Each point uses 2,500 simulated experiments and 999 treatment reallocations. Results are reproducible from `analysis/power_simulation.R`.

Table 3: Approximate completed sample required for 80% power

Recovery effect	10 points	15 points	20 points	25 points	30 points
Completed participants	1,520	740	460	340	260

Thresholds use the central assumptions in panel (a) and are interpolated from the monotone curves, then rounded up to the next 20 participants. They are conditional benchmarks rather than universal minimum sample sizes.

At 300 completed participants, the central simulation yielded 61% power for a 20-point effect, 78% for 25 points, and 91% for 30 points. The smallest effect reaching 80% power at this sample was approximately 26 percentage points. This value remained between 25.6 and 25.9 points across the permitted 28-, 29-, and 30-pool configurations. Conversely, a 20-point effect required approximately 460 participants under the central assumptions. Its 80% threshold moved from 360 participants under higher eligibility and lower clustering to 640 under lower eligibility and higher clustering. A higher persistence reversal rate produced a threshold of 540. The critical sample size was therefore conditional on the unknown behavioral parameters. Given the fixed budget, the simulation showed most directly that the study could reliably detect large changes in sustained reversal, while smaller effects would remain imprecisely estimated.

The primary randomization test reproduced the realized paired and unmatched assignments. Regression estimates clustered uncertainty by pool, adjusted for the prespecified pool-size indicator, and used small-sample inference. An equal-pool-weight analysis checked whether results depended on larger pools contributing more participants. Each oTree session remained a separate interaction and randomization cohort.

Strategic decisions did not time out automatically. The software recorded response time and flagged decisions exceeding 90 seconds, but it never generated a ballot or allocation. The researcher or a trained research assistant monitored the session and prompted delayed participants to complete the current decision so the group could advance.

6 Implementation, Ethics, and Project Timeline

Implementation followed the approved protocol at UTU Choice Lab. Adults registered through UTU RPP/TOKA, which managed recruitment and attendance separately from the experiment. Before practice, oTree presented study-specific information and recorded consent. Participants learned that the detailed research question would be explained after the paid task because advance disclosure could influence the institutional ballot. All task rules, payments, data processing, foreseeable risks, and the right to stop were disclosed beforehand. The full research purpose was explained immediately after paid decisions.

The researcher or a trained research assistant fixed a consenting roster of 10, 15, or 20 before paid play. They formed the matching pools in Table 1 and monitored synchronized progress. No participant entered or changed pools after paid play began. A technical pilot verified rematching, both Block-2 conditions, wait pages, response-time flags, browser recovery, data exports, and payments. A behavioral pilot assessed comprehension, duration, fatigue, choice distributions, and expected earnings. The protocol, analysis plan, and data dictionary were then frozen before confirmatory collection.

Live oTree data were hosted on University of Turku IT servers in Finland. The public interface did not collect research data, and oTree used no artificial-intelligence functions. UTU RPP/TOKA received no oTree codes, experimental responses, or payment information. The research team made

immediate payments using a temporary code-and-amount list. Any named receipt contained no oTree code. The temporary list was destroyed after payment reconciliation and no later than 30 November 2026. The retained research dataset was then anonymized. Before anonymization, access was limited to Janne Tukiainen, Hector Bahamonde, and authorized personnel.

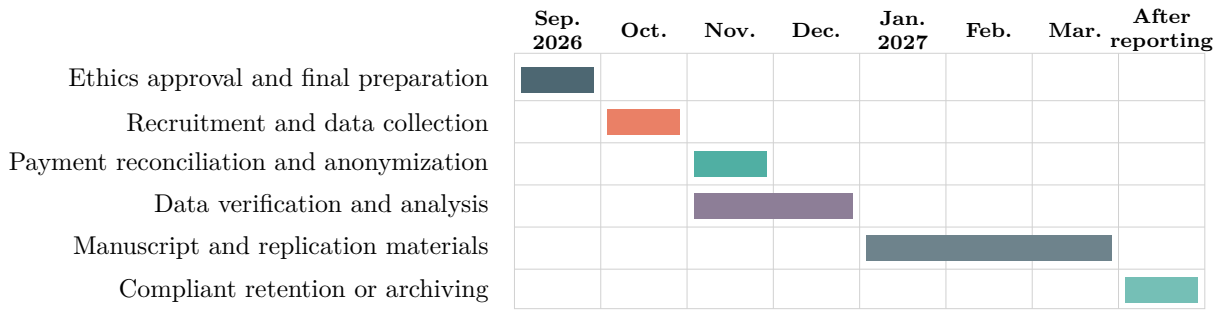


Figure 7: Study timeline under the approved protocol.

Notes: Data collection took place from 1–31 October 2026 at UTU Choice Lab. Payment reconciliation and destruction of the temporary code-and-amount list were completed no later than 30 November 2026. Reporting, disclosure review, and retention or archiving followed the approved data-management plan.

The study produced academic articles, presentations, and replication materials. Consistent with University open-science policy, documentation, code, and data that could be shared safely were made as open as possible and as restricted as necessary.

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